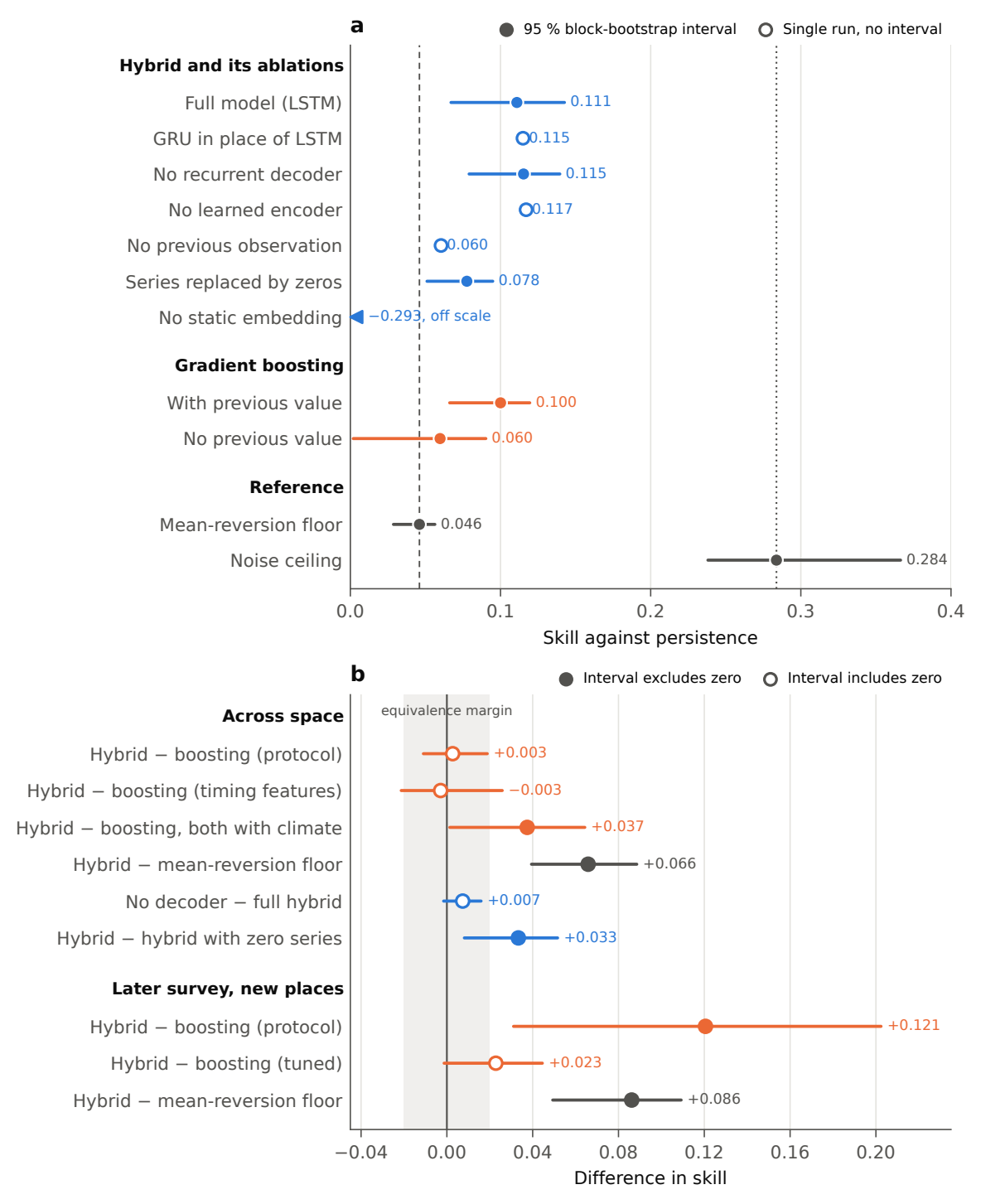
a
95 % block-bootstrap interval
Single run, no interval
Hybrid and its ablations
Full model (LSTM)
0.111
GRU in place of LSTM
0.115
No recurrent decoder
0.115
No learned encoder
0.117
No previous observation
0.060
Series replaced by zeros
0.078
No static embedding
−0.293, off scale
Gradient boosting
With previous value
0.100
No previous value
0.060
Reference
Mean-reversion floor
0.046
Noise ceiling
0.284
0.0
0.1
0.2
0.3
0.4
Skill against persistence
b
Interval excludes zero
Interval includes zero
Across space
equivalence margin
Hybrid − boosting (protocol)
+0.003
Hybrid − boosting (timing features)
−0.003
Hybrid − boosting, both with climate
+0.037
Hybrid − mean-reversion floor
+0.066
No decoder − full hybrid
+0.007
Hybrid − hybrid with zero series
+0.033
Later survey, new places
Hybrid − boosting (protocol)
+0.121
Hybrid − boosting (tuned)
+0.023
Hybrid − mean-reversion floor
+0.086
−0.04
0.00
0.04
0.08
0.12
0.16
0.20
Difference in skill